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Rural–Urban Commuting Area Codes – Documentation

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Background

The Rural-Urban Commuting Area (RUCA) codes are a detailed and flexible scheme for delineating sub-county rural and urban areas using census tracts. The RUCA codes were designed to address a major limitation associated with county-based rurality classifications. Counties (and associated county equivalents) are often too large to accurately delineate boundaries between rural and urban areas. In fact, most counties include people living in both rural and urban territory. This is especially true for the large counties of the western United States. Therefore, the more geographically detailed classification provided by the RUCA codes affords a more accurate distinction between rural and urban areas. This, subsequently, improves research and policy focused on rural residents and communities.

The RUCA codes are based on the same theoretical concepts used by the U.S. Office of Management and Budget (OMB) to define county-level metropolitan and micropolitan areas. Central place theory asserts that there is a hierarchy of urban locations, with larger cities providing a wider variety of specialized goods and services (e.g., neurosurgeons, health food or ethnic grocery stores) and smaller cities only providing more basic goods and services (e.g., primary care physicians, convenience stores). People living in rural areas travel to the closest city to purchase the goods and services they need. And if the closest city doesn't have what they need, they go to the closest larger city that has it.

This connection between rural and urban areas is not limited to purchasing goods and services; the connection also affects employment. The closer a rural resident is to a city, the more likely he or she is to be employed in that city. Areas with larger shares of the population commuting to an urban core are considered more economically tied to that core. To capture this connection, OMB uses information on the size of urban areas and commuting flows to identify metropolitan and micropolitan statistical areas.

We apply similar, though not identical, criteria to measures of urbanization and commuting for census tracts to identify urban cores and adjacent territory that is economically integrated with those cores. We adopt OMB's metropolitan and micropolitan terminology to highlight the underlying connectedness between the two classification systems. However, the use of census tracts instead of counties provides a different and more detailed geographic pattern of boundaries within the rural-urban hierarchy.

These hierarchical consumption and employment patterns are the reason for the two levels of RUCA codes. The primary RUCA codes establish the urban cores and the census tracts that are the most economically integrated with them. Each of the 10 primary RUCA codes indicates whether a census tract is part of an urban core or connected to it through commuting, as well as the size of the urban core and the strength of the commuting connection. Based on these primary relationships, each census tract is then assigned a secondary RUCA code. The 21 secondary RUCA codes indicate whether a census tract has a strong secondary connection (through commuting) to an urban core higher up the urban hierarchy. Few, if any, research or policy applications need the full set of codes. However, the system allows for the selective combination of codes to meet varying definitional needs. See the [Users' Guide](#) for more information and examples.

Data Sources

To create the census tract RUCA codes, three data inputs are needed: total population for census tracts and blocks, the most recent urban area delineation, and commuting flows. All three of these inputs come from the U.S. Department of Commerce, Bureau of the Census (Census Bureau), but they are released at different times and use different surveys.

Population and urban area data were tabulated from the [2020 decennial Census of Population and Housing](#) [🔗](#). Census tracts are geographic areas used by the Census Bureau when collecting data. They have an optimal size of 4,000 individuals, producing small census tracts in densely settled urban locations and very large census tracts in sparsely settled rural locations. The Census Bureau creates urban areas from census blocks (the smallest collection unit) that meet a set of population density, size, and built environment criteria. Urban areas encompass residential, commercial, and other non-residential land uses. We used data on the total population of census tracts and urban areas, as well as block-level urban population to construct the RUCA codes.

Tract-to-tract commuting flows were tabulated using the 2017–21 American Community Survey (ACS), as part of a special tabulation for the U.S. Department of Transportation—the [Census Transportation Planning Package \(CTPP\)](#) [🔗](#). Census tracts are the smallest geographic area for which the CTPP data are available. The ACS is collected annually from an initial sample of more than 3.5 million housing units and

was combined across 5 years (2017–21) to create a large enough sample to produce census-tract estimates. As with all survey data, ACS estimates are not exact because they are based on a sample. In general, the smaller the sample, the larger the degree of uncertainty associated with it.

To create the ZIP code RUCA classification, we use the census tract RUCA codes in conjunction with two additional data sources. The first is ZIP code [areas](#) [↗](#) and [points](#) [↗](#) (accessed April 2025), curated by ESRI using June 2024 data from TomTom North America, Incorporated and the U. S. Postal Service . The other data source is 3-arc-second gridded 2020 nighttime population data from the Oak Ridge National Laboratory’s [LandScan USA](#) [↗](#) dataset.

Scope/Coverage of Data

The 2020 census tract RUCA codes data and June 2024 ZIP code RUCA code data are available for the United States and its territories. CTPP commuting flow data are not available for U.S. territories, so RUCA codes were assigned in these locations based solely on urbanization.

Methods

Census Tract Methodology

The census tract RUCA codes are created in three steps. In the first step, census tracts are grouped into urban cores, leaving rural tracts ungrouped. To group census tracts into urban cores, we aggregate the census block urban population to obtain an urban population for each census tract. Then, we calculate the share of each census tract’s residents living in an urban area. If at least half of the residents are living in a single urban area, the census tract is considered part of an urban core and is assigned to that urban area. Otherwise, the tract is considered rural.

These urban cores approximate the U.S. Census Bureau’s urban areas using census tracts. The cores are classified as metropolitan, micropolitan, or small town, based on the total population of their associated urban area. Each urban core, and the census tracts that comprise it, is considered a single statistical unit

going forward in the classification process; however, rural census tracts continue to be considered independently.

In the second step, the largest and second largest commuting flows among rural census tracts and urban cores are identified. Using the tract-to-tract commuting flow data, total workers commuting are aggregated by residential origin and workplace destination. The two highest commuting flows are identified and the share of commuters travelling to the top two destinations is calculated. If two commuting flows from an origin are the same size, the destination with the larger population is chosen.

In the last step, the primary and secondary RUCA codes are calculated.

Primary RUCA Codes

The 10 primary RUCA codes shown in the table below are created by combining an urban core with the primary, or largest, commuting flow. Urban cores are classified into one of three groups by size. Metropolitan cores (code 1) are census tract equivalents of the urban areas at the center of Metropolitan Statistical Areas. Micropolitan cores (code 4) are the census tract equivalents of the urban areas at the center of Micropolitan Statistical Areas. Small town cores (code 7) are the census tract equivalents of all other urban areas. For all three urban core RUCA codes, the primary commuting flow is within the urban core.

High commuting shares (codes 2, 5, and 8) indicate a primary commuting flow of at least 30 percent to an urban core. These are generally tracts that were not included in an urban core. However, many micropolitan and small town cores themselves (and even a few metropolitan cores) have a high enough primary commuting flow to another urban core to be considered high commuting. Typically, these areas are not job centers themselves but serve as bedroom communities for a nearby, larger city. In this case, the areas are not classified as an urban core in the RUCA codes but a high commuting share location to the larger urban core. Low commuting shares (codes 3, 6, and 9) indicate a primary commuting flow of 10 to 30 percent to an urban core.

These commuting-share codes identify nearby territory or "influence areas" of metropolitan, micropolitan, and small town cores and are similar in concept to the "nonmetropolitan adjacent" codes

found in other ERS rural-urban county classification schemes ([Rural-Urban Continuum Codes](#), [Urban Influence Codes](#)).

Finally, rural census tracts (code 10) are not part of an urban core and the primary commuting flow is local or to another rural tract. In a few cases, a rural census tract's primary commuting flow is to an urban core but is less than 10 percent of all commuters. In these cases, the tract is not considered fundamentally connected to an urban core and is considered rural.

Primary RUCA codes, 2020

Code	Classification description
1	Metropolitan core: primary flow is within an urban area of 50,000 or more people (metro UA)
2	Metropolitan high commuting: primary flow is 30 percent or more to a metro UA
3	Metropolitan low commuting: primary flow is 10 percent to 30 percent to a metro UA
4	Micropolitan core: primary flow is within an urban area of 10,000 to 49,999 people (micro UA)
5	Micropolitan high commuting: primary flow is 30 percent or more to a micro UA
6	Micropolitan low commuting: primary flow is 10 percent to 30 percent to a micro UA
7	Small town core: primary flow is within an urban area of 9,999 or fewer people (small town UA)
8	Small town high commuting: primary flow is 30 percent or more to a small town UA
9	Small town low commuting: primary flow is 10 percent to 30 percent to a small town UA
10	Rural area: primary flow is to a tract outside an UA

Code	Classification description
99	Not coded: census tract is entirely in coastal or inland water bodies, with zero population and zero land area
UA = urban area; primary flow = largest commuting flow from a census tract Source: USDA, Economic Research Service	

Secondary RUCA codes

The primary RUCA codes offer a relatively straightforward and complete delineation of metropolitan and nonmetropolitan areas based on the size and direction of primary commuting flows. However, the second largest, or secondary, commuting flows may indicate other connections among rural and urban places. We incorporate these other connections into the secondary RUCA codes.

The secondary RUCA codes are created by subdividing some of the primary RUCA codes to identify areas where classifications overlap, based on the size and direction of the secondary commuting flow (table below). The secondary codes indicate a fundamental connection to a larger urban core further up the urban hierarchy. For example, the secondary RUCA codes 4.1 and 5.1 identify areas where the primary commuting flow is within or to a micropolitan core, but at least another 30 percent of commuters travel to a workplace in a metropolitan core. Similarly, codes 10.1, 10.2, and 10.3 identify rural tracts for which the primary commuting share is local, but 30 percent or more of that tract's commuters also travel to a nearby metropolitan, micropolitan, or small town core for work, respectively.

Secondary RUCA codes, 2020

Code	Classification description
1 Metropolitan core: primary flow is within an urban area of 50,000 or more people (metro UA)	
1.0	No additional code
1.1	Secondary flow is 30 percent to 50 percent to a larger UA

Code	Classification description
2 Metropolitan high commuting: primary flow is 30 percent or more to a metro UA	
2.0	No additional code
2.1	Secondary flow is 30 percent to 50 percent to a larger UA
3 Metropolitan low commuting: primary flow is 10 percent to 30 percent to a metro UA	
3.0	No additional code
4 Micropolitan core: primary flow is within an urban area of 10,000 to 49,999 (micro UA)	
4.0	No additional code
4.1	Secondary flow is 30 percent to 50 percent to a metro UA
5 Micropolitan high commuting: primary flow is 30 percent or more to a micro UA	
5.0	No additional code
5.1	Secondary flow is 30 percent to 50 percent to a metro UA
6 Micropolitan low commuting: primary flow is 10 percent to 30 percent to a micro UA	
6.0	No additional code
7 Small town core: primary flow is within an urban area of up to 9,999 (small town UA)	
7.0	No additional code
7.1	Secondary flow is 30 percent to 50 percent to a metro UA
7.2	Secondary flow is 30 percent to 50 percent to a micro UA
8 Small town high commuting: primary flow is 30 percent or more to a small town UA	

Code	Classification description
8.0	No additional code
8.1	Secondary flow is 30 percent to 50 percent to a metro UA
8.2	Secondary flow is 30 percent to 50 percent to a micro UA
9 Small town low commuting: primary flow is 10 percent to 30 percent to a small town UA	
9.0	No additional code
10 Rural area: primary flow is to a tract outside an UA	
10.0	No additional code
10.1	Secondary flow is 30 percent to 50 percent to a metro UA
10.2	Secondary flow is 30 percent to 50 percent to a micro UA
10.3	Secondary flow is 30 percent to 50 percent to a small town UA
99 Not coded: Census tract contains only water and has zero population	
UA = urban area; primary flow = largest commuting flow from a census tract; secondary flow = second largest commuting flow from a census tract Source: USDA, Economic Research Service	

ZIP Code Methodology

While the RUCA codes are created using census tracts, researchers and policymakers find it useful to have an approximation for ZIP codes. ZIP codes are used by the U.S. Postal Service to process and deliver mail more quickly and efficiently. Therefore, ZIP codes do not necessarily nest within or consist of U.S. Census Bureau collection units such as census tracts. They also change frequently as new postal routes are created, post offices are closed, or other changes occur.

To create this second dataset, we transfer the 2020 RUCA codes from census tracts to ZIP codes. The ZIP code version of the RUCA code data product includes both "area" ZIP codes and "point" ZIP codes, the latter representing post offices, office buildings, and other large volume customers. The transfer of tract-based RUCA classifications to ZIP code areas was carried out using ESRI ArcGIS spatial analysis tools on census tract and ZIP code area boundary files, gridded population data, and the census tract RUCA codes. The methodology differs for "area" and "point" ZIP codes, due to their different physical areas.

Area ZIP code methodology

Many ZIP code areas overlap multiple census tracts. If the census tracts associated with a ZIP code area all had the same tract-based RUCA code, that RUCA code was assigned to the ZIP code area. However, if the census tracts associated with a ZIP code area had different RUCA codes, the RUCA code with the highest population was assigned to the ZIP code. To transfer the census tract RUCA codes to ZIP code areas:

1. We combined the census tracts with the same secondary RUCA codes into a single geographic area. This removes census tract borders between adjacent census tracts with identical secondary RUCA codes. (Secondary RUCA codes are a subdivision of primary RUCA codes, so they retain the information needed to assign both primary and secondary RUCA codes to ZIP codes.)
2. We overlaid the ZIP code areas and secondary RUCA code areas and split each ZIP code into separate pieces that fall into different secondary RUCA code areas. The result is a set of combined ZIP/RUCA code areas.
3. We aggregated the gridded population data within each ZIP/RUCA code area to produce a population estimate for each secondary RUCA code within a ZIP code. A population grid is included in a ZIP/RUCA code area's total if its center point (centroid) is within the area's boundaries.
4. We calculated the share of the ZIP code area's total population living within each secondary RUCA code area and used the population shares to assign primary and secondary RUCA codes to the entire ZIP code area.
 - a. First, the primary RUCA code with the largest share of the ZIP code's population was assigned to the ZIP code area. The share of a ZIP code's population living in a primary RUCA code is the

sum of the shares in all secondary RUCA codes with the same integer (e.g., primary RUCA code 7 = secondary RUCA codes 7.0 + 7.1 + 7.2).

- b. Then, of the secondary RUCA codes derived from the assigned primary RUCA code, the one with the highest population share within the ZIP code was assigned. (e.g., If the ZIP code area's primary RUCA code was 7, and the share of the ZIP code's population living in secondary RUCA code 7.1 was larger than the shares living in secondary RUCA codes 7.0 and 7.2, the ZIP code's secondary RUCA code would be 7.1.)
- c. When assigning the RUCA codes, if the ZIP code area's population was spread equally through the options, the RUCA code with the largest area was selected.

Three ZIP code areas could not be assigned using the above method due to their small size. Each was only associated with one primary and second RUCA code, which they were subsequently assigned.

Point ZIP code methodology

The ZIP code "point" data were used to identify ZIP codes that only exist as points, such as post offices or large volume customers. Transferring census tract RUCA values to these types of ZIP codes does not require a population analysis, as a point can only fall into one census tract. We used the "identity" tool in ArcGIS to identify the census tract corresponding to each of these ZIP code points. We then assigned the corresponding census tract's primary and secondary RUCA codes to the ZIP code points.

Strengths and Limitations

The USDA, ERS's Rural-Urban Commuting Area (RUCA) codes were designed to more accurately delineate boundaries between rural and urban areas than can be done with counties. The data classify census tracts using a flexible, two-level scheme that allows researchers and policymakers to create a model of the relationships among differing sized urban areas, their economically connected neighbors, and more isolated rural areas that suits their needs. Therefore, the RUCA codes provide a more geographically detailed, sub-county classification to aid research and policy focused on rural residents and communities.

RUCA codes corresponding to each decennial census since 1990 are available, providing a consistent source of sub-county delineation of rural and urban areas. However, RUCA codes are not directly comparable over time due to a combination of changes in the underlying data and changes to the method used to construct them. For instance, census tracts are reconfigured each decade, making it difficult to compare them over time. There have also been changes to how the Census Bureau delineates urban areas that decreased rural population and territory in 2000 but increased them in 2020. Between 2000 and 2010, the data source for commuting patterns also switched from the decennial census in 2000, which measured one point in time, to the American Community Survey in 2010, which provided a 5-year average during 2006–10. Despite these changes, the RUCA codes have always been created using the best available data, resulting in their broad adoption across policy and research applications.

For the 2020 RUCA codes, we made a methodological change that impacted how an urban core was defined. For the 2010 RUCA codes and earlier, a census tract was considered part of an urban core if more than 30 percent of the tract's residents lived in a single urban area or if at least half of its residents lived in any urban area. For the 2020 RUCA codes, a census tract was considered part of an urban core only if at least half the tract's residents lived in a single urban area.

While census tracts are the smallest sub-county unit that can be used to create the RUCA codes, they are imperfect units, especially in rural areas. Because census tracts are based on an ideal population size, they can vary greatly in area. This means that in sparsely populated areas, tracts can be so large that they are coextensive with counties. And their population is not spread evenly throughout the area. So, a community economically tied to an urban core that is located in one corner of the tract can affect the classification of the entire tract. This phenomenon is similar to what occurs in counties, but at a smaller scale.

Another limitation of the RUCA codes is associated with the data inputs used to create them. First, the Census Bureau changed the criteria used to delineate urban areas between 2010 and 2020, changing the minimum population threshold from 2,500 residents to 5,000 residents and adding an alternative minimum of 2,000 housing units. The result of this change is that there are fewer small town cores and associated commuting areas in the vintage 2020 RUCA data.

Second, there are also several data issues affecting the 2017–21 Census Transportation and Planning Package (CTPP) estimates. In accordance with data privacy requirements, the Census Bureau does not

provide commuting flow estimates when there are fewer than three observed trips between two census tracts. The Census Bureau has also perturbed the data and rounded estimates to the nearest 5 trips to protect the privacy of survey responders (e.g., 8 to 12 commuters is rounded to 10).

There are some challenges in the underlying survey responses as well. If the workplace destination is unclear in the survey response, the trip is omitted. Due to limited resources to deduce unclear workplace destinations, this is the case for nearly 25 percent of workers. Additionally, the COVID-19 pandemic started during the sample period, affecting the commuting behaviors of workers toward the end of the sample period. During lockdown, data collection for the American Community Survey was suspended, reducing its effect on the data; however, the changes in commuting patterns due to the COVID-19 pandemic lasted through the end of the sample period, affecting the CTPP commuting flow estimates.

Finally, the margins of error, which indicate the confidence of the estimate, can be larger than the estimated commuting flow itself for some tracts. This situation generally occurs due to a small number of survey responses, which is more common in less populated areas.

Despite these data challenges, CTPP is the best data available for measuring tract-to-tract commuting flows and the large margins of error in the data have minimal impact on a tract's RUCA codes classification. For more information on the methods used to create the CTPP data, refer to the [CTPP Methodology Report](#) [↗](#). For details on the work analyzing the impact of margins of error on RUCA code classification, refer to [Fowler and Cromartie \(2023\)](#) [↗](#).

Related Data / Data Linkages

To determine a census tract, please see the Federal Financial Institutions Examination Council's [FFIEC Geocoding/Mapping System](#) [↗](#).

Recommended Citation

U.S. Department of Agriculture, Economic Research Service. *2020 Rural-Urban Commuting Area Codes*, July 2025.



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